

Assignment 8:

Ganesh Gundekarla

11700551

Summary:

a. Data generation by using samples

Here a sampled data is being done and generated using the make blobs function that is being available on the scikit learn library, the goal of this function is to make a points i.e the cluster set of the points which are based on specific centers and standard deviation with some random state applied to it .

b. Visualization of the data

So, the above randomly generated data is being generated is being plotted using a scatter plot which can be used as a reference to do check it with the kmeans output , this scatter plot shows the distribution of the samples spread across the space.

c. K means clustering: the kmeans is then applied on the generated data using the class kmeans , here we specify the no of the clusters as 4 which is very important .

d. The fit method is then used to fit in the data meaning model the data , which then the predict method is used to assign the data point to one. Of the clusters which is the nearest one.

c. Visualization on the resulted data:

so, the data which is being grouped is then visualized using the same scatter plot and each point is assigned to a cluster using the k means algo and the color is assigned to show where the cluster is present.

Code :

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobss
from sklearn.clusterr import KMeanss

# Generate sample data using random values with a specifid set of
conditions
variable_temp, _ = make_blobss(n_samples=300, centers=4,
cluster_std=0.60, random_state=0)

# Visualize the data aka original data visualization
plt.scatter(variable_temp[:, 0], variable_temp[:, 1], s=50)
plt.xlabel('feature on x axis')
plt.ylabel('feature on y axis')
plt.title('Original Data that is generated')
plt.show()

# Apply k-means clustering on the group of points
kmeanss = KMeanss(n_clusters=4)
kmeanss.fit(variable_temp)
y_kmeanss = kmeanss.predict(variable_temp)

# Visualize the clustered data after the set of points.
plt.scatter(variable_temp[:, 0], variable_temp[:, 1], c=y_kmeanss,
s=50, cmap='viridis')
centers = kmeanss.cluster_centers_
plt.scatter(centers[:, 0], centers[:, 1], c='red', s=200, alpha=0.5)
plt.xlabel('x axis ')
plt.ylabel('y axis')
plt.title('Clustered Data after kMeans clustering done ')
plt.show()
```




